0 1 2 3 4 5 6 7

arr: [-5, -3, -1, 6, -2, -5, 5, 3]

0 1 2 3 4 5 6 7

bool: [T T T F T T F F]

TC: O(n).

AS: O(1)

[0, -10, 1, 3, -20]

0 1 2 3 4

[-7, 7, -1, 3, 7]

0 1 2 3

[-2, -1, 3, 5]

0 1 2 3

[-2, -1, -3, 5]

(n × m).

X0	0	X0
1	X0	1
1	X0	1

X2	0	X2
1	X2	1
1	X2	1

⇒

0	0	0
1	0	1
1	0	1

TC = O((nm)(n+m))

X2	0	X2
1	X2	1
X2	0	X2

1	1	1	1
1	0	1	1
1	1	1	0
1	1	1	1

	0	1	2
0	X0	0	X0
1	1	X0	1
2	X0	0	X0

rows: | T | F | T | → O(n)

cols: | F | T | F | → O(m)

TC = O(nm + nm) = O(nm).

AS = O(n + m).

1	1	1	1	1
1	0	1	1	1
1	1	1	0	1
1	1	1	1	0

n = 4

m = 5.

1	X0	1	0X	0X
0X	0	X0	X0	X0
0X	X0	X0	0	X0
0X	X0	X0	X0	0

⇒

1	0	1	0	0
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0

1	1	1	1	1
1	1	1	1	1
1	1	0	1	1
1	1	1	1	0

1	1	X0	1	X0
1	1	X0	X0	X0
0X	X0	0	X0	X0
0X	X0	X0	X0	0

→

1	1	0	1	0
1	1	0	0	0
0	0	0	0	0
0	0	0	0	0

TC = O(nm)

AS = O(1).

1	1	0	1	1
1	1	1	1	1
1	1	1	1	1
0	1	1	0	1

0	0	0	0	0
0	1	0	0	1
0	1	0	0	1
0	0	0	0	0

1	1	0	X0	1
1	1	X0	X0	1
1	1	X0	X0	1
0	1	X0	X0	0

⇒

1	1	0	0	1
1	1	0	0	1
1	1	0	0	1
0	0	0	0	0

[bool first_row_zero : true.

bool first_col_zero : true.]